Minor Project – (V Sem)
Mid Phase-1 Project Evaluation – 2 Credits (CIE-100 M)
Project Presentation & Activity Report
Course Code : BIC586

Rajarajeswari College of Engineering, Bangalore, Karnataka
Department of Computer Science & Engineering (IoT CS including BCT)

Title of the Minor Project Work – Write here

Minor Project
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Semester : V

Project Guide : Prof. Bhagyalakshmi.P.T

Date of review
20 Nov 2025

Overview / Flow of the Minor Project Presentation

1. Introduction
2. Motivation obtained to do the project
3. Aim of the Project work & Problem Statement with Objectives
4. Literature Review
5. Methodology adopted with Block-Diagrams & Flow- Charts
6. H/w and S/w tools used
7. Working of the entire project
8. Results (Simulation & Experimental) – Photographs & Snapshots
9. Applications, Advantages, Outcome and Limitations
10. Conclusion and Future Work
11. Video of the entire working module
12. Reflections
13. References
14. Publications/Any other recognitions / awards, prizes if any-achievements

Brief Introduction to the project work

1. Introduction

The Smart Automation System is an IoT-based solution designed to automate and control electrical loads in homes, industries, farms, and commercial buildings. It uses ESP32 microcontrollers, Solid State Relays (SSRs), and sensor-driven intelligence to provide real-time monitoring, safety automation, and efficient remote control.
The system aims to make infrastructure smarter, safer, and more energy-efficient.

Motivation obtained to take up the project with problem definition

- Increasing need for remote automation in homes and industries.
- High failure rate of mechanical relays → need for safe solid-state switching.
- Growing importance of IoT in smart cities.
- Need for real-time safety response to fire, smoke, and water leak incidents.
- Desire for a low-cost and scalable automation solution that can be easily deployed anywhere.

Aim of the Project work with 3 Objectives

❑ Aim

- To develop a smart IoT-based automation system using ESP32 and SSR to safely control and monitor electrical loads remotely with integrated emergency response features.

❑ Problem Statement

- Existing automation systems are:
- Expensive
- Limited in scalability
- Lack safety automation
- Not suitable for industrial-level reliability

❑ Objectives

- To design an IoT controller using ESP32 with real-time Wi-Fi connectivity.
- To implement SSR-based safe switching for appliances.
- To integrate sensors for fire, smoke, water, and environment monitoring.
- To develop a user-friendly mobile/desktop interface.
- To design emergency shutdown mechanisms.
- To test the system in home, industrial, and agricultural scenarios.

Literature Review / Survey / Gaps

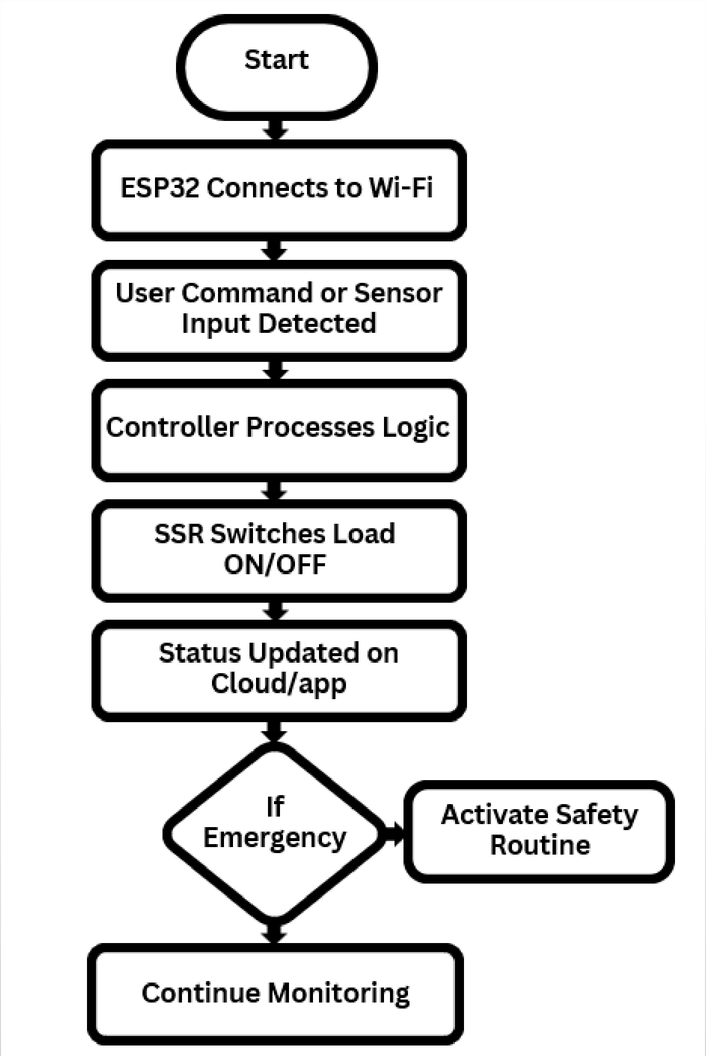
- **Research on IoT automation shows increasing use of Wi-Fi microcontrollers for smart home systems.**
- **Solid State Relays are preferred over electromechanical relays due to fast switching and long lifespan.**
- **Studies suggest automation improves energy efficiency by 20–30%.**
- **Emergency automation significantly reduces electrical fire accidents.**
- **ESP32 is widely used for IoT due to its low power consumption, dual-core CPU, and built-in connectivity.**

Proposed Methodology adopted with Block-Diagrams & Flow- Charts / DFD / Algorithms

❑ Block Diagram Components

- User App / Dashboard
- Cloud or Wi-Fi Router
- ESP32 IoT Controller
- Solid State Relay Board
- Sensors (Smoke, Fire, Temperature, Water Leak)
- Electrical Loads (Lights, Fans, Pumps, Motors)

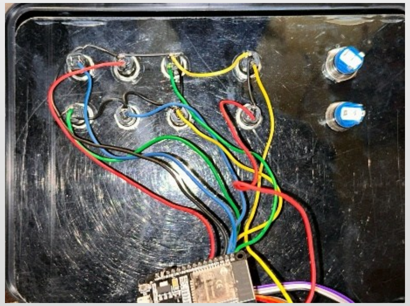
❑ Flow Chart:



H/w and S/w tools used

HARDWARE COMPONENTS:

- ESP32 Development Board
- Solid State Relays (SSR)
- DHT11/22 Temperature Sensor
- Smoke/Fire Sensor (MQ-series)
- Water Leak Sensor
- Power Supply 5V/12V
- Supporting circuit modules



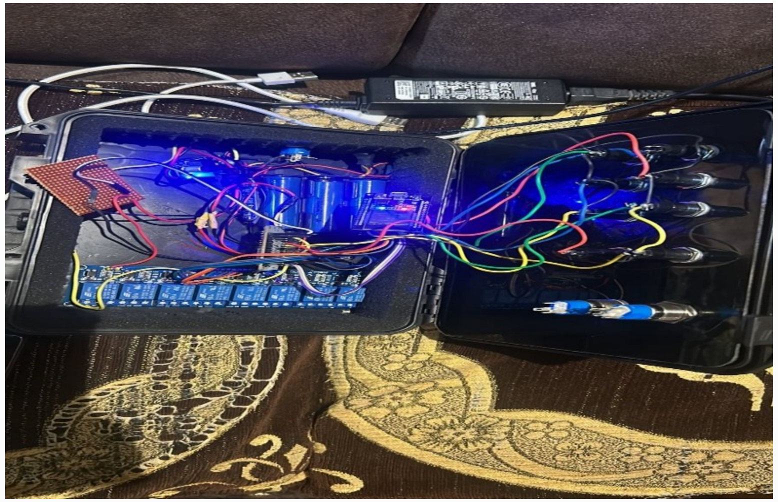
SOFTWARE COMPONENTS:

- Arduino IDE / ESP-IDF
- Firebase/MQTT/HTTP server
- Mobile App / Web Dashboard
- OTA update software
- Data logging system

Working of the entire project (proposed - tentative)

- **Working of the Entire Project**
- **ESP32 connects to Wi-Fi and links to cloud/app.**
- **Users send commands through mobile app or dashboard.**
- **Controller receives commands and triggers SSR to switch loads.**
- **Sensors continuously monitor safety conditions.**
- **In case of smoke/fire/water leak → automatic emergency shutdown.**
- **Real-time status is displayed to the user.**
- **Recorded logs help monitor usage and safety events.**

Results (Simulation & Experimental) – Photographs & Snapshots (expected)



- **Simulation Results:**

  - **Successful Wi-Fi connectivity**
  - **Correct switching logic in software**
  - **Emergency rule triggers properly**

- **Experimental Results:**

  - **SSR switching works without noise**
  - **ESP32 responds instantly**
  - **Auto-shutdown activates during smoke/water detection**
  - **Real-time dashboard shows accurate data**

Conclusion and Scope for Future Work

- **CONCLUSION:**

The Smart Automation System successfully automates electrical loads, improves safety, and provides scalable IoT control. The use of ESP32 and SSR ensures reliability, low power consumption, and fast response. Safety sensors add protection for real-world use.

FUTURE WORK:

- **AI-based automation**
- **Voice assistant integration**
- **LoRaWAN for long range**
- **Smart energy analytics**
- **Multi-user access control**
- **Fully polished mobile application**

Applications, Advantages, Usefulness to Society, Outcome and Limitations

- Turns appliances ON/OFF remotely.
- Automates lights, fans, and other devices.
- Provides safety by preventing overload.
- Saves electricity with timers and scheduling.
- Supports voice control (Alexa/Google).
- Useful for home, office, and industrial automation
- Helps in security with sensors and alerts.

References used (base papers) & Publications (if any done)

- **References:**
  - ESP32 Technical Documentation
  - IEEE Research Papers on Home Automation
  - Solid State Relay Datasheets
  - IoT Protocol Documentation (MQTT/HTTP)
  - Arduino & ESP-IDF Libraries

- **Publications / Awards / Achievements:**
  - Project presented in college technical exhibition.
  - Selected for campus innovation showcase.

Reflections – Project Flow Line

April – 2025 :	
May - 2025 :	
June - 2025 :	
July - 2025 :	
Aug - 2025 :	
Sept - 2025 :	
Oct - 2025 :	
Nov - 2025 :	
Dec - 2025 :	

Here, you list out the Technical and Non-technical things that you learnt during this major-project


Thank
You


Questions
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